Simulation Report

=== SIGNAL DETECTION ANALYSIS REPORT ===

1. File Statistics:
- Total files analyzed: 6
- Periodic signal files: 4
- Random signal files: 2

2. Period Detection Results:
- Total true periods: 4
- Total detected periods: 4
- Correctly detected true periods: 2

3. File-level Detection Accuracy:
- Files with strictly correct period detection: 2
- Files with relaxed correct period detection: 2
- Correctly rejected random signals: 1

4. Performance Metrics:
- Detection rate: 50.00%
- False positive rate: 50.00%
- Strict accuracy for periodic signals: 50.00%
- Relaxed accuracy for periodic signals: 50.00%
- Accuracy for random signals: 50.00%
- Overall accuracy: 50.00%

Source Name	Signal Type	Real Period	Detected Period	Detected Period Error	Judgment
1_periodic245.71days	periodic_signal	245.71	245.8128	2.5236	True
2_periodic526.05days	periodic_signal	526.05	525.303	0.918	True
3_quasiperiodic1410.01days	periodic_signal	1410.01	-	-	False
4_quasiperiodic1438.46days	periodic_signal	1438.46	-	-	False
5_random001.txt	random_signal	-	-	-	True
6_random002.txt	random_signal	-	-	-	False

Analysis Parameter Settings

lsp_params

Parameter Name	Parameter Value
M	10000
n_jobs	10
multiple_freq_max	100
divide_freq_step	10

lsp_filter

Parameter Name	Parameter Value
height	0.1
min_prominence_rate	0.5
self_harmonic	True
reverse	True
n_harmonics	2
sigma_threshold	2.0
peak_width_factor	0.1
snr_threshold	5.0
fap_threshold	0.001

dcf_params

Parameter Name	Parameter Value
delta_tau	30
c	40
max_tau	3000
normalize	True

dcf_filter

Parameter Name	Parameter Value
height	0.15
prominence	0.3
snr_threshold	3.0
distance_rate	4
self_harmonic	True
sigma_threshold	2.0
reverse	True

jv_params

Parameter Name	Parameter Value
test_periods_start	100
test_periods_end	3000
test_periods_step	10
m_bins	2

jv_filter

Parameter Name	Parameter Value
max_serise	3000
v2_threshold	0.2
min_peak_distance	10
prominence	0.1
self_harmonic	True
sigma_threshold	2.0
reverse	True

wwz_params

Parameter Name	Parameter Value
c	0.0125
scan_range	0.8
freq_step	0.0001
tau_number	400
z_height	20000

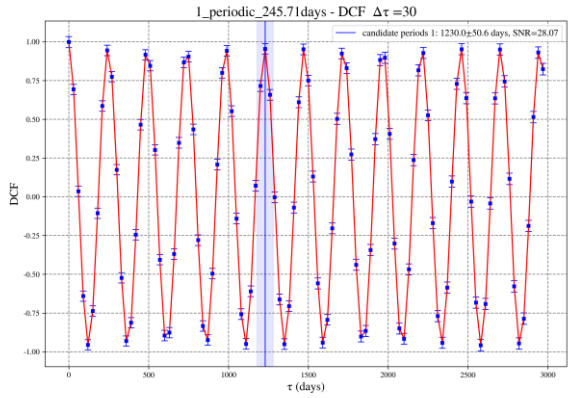
wwz_filter

Parameter Name	Parameter Value
min_size	10
confidence	0.99
peak_mode	peak

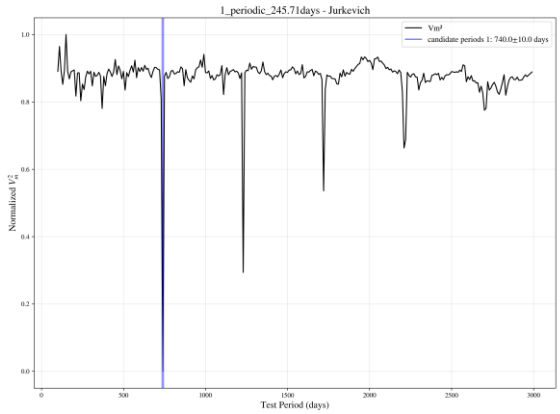
Image Groups

1_periodic245.71days

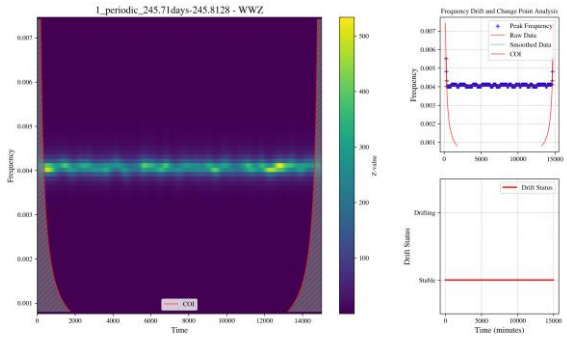
LSP



DCF



JV

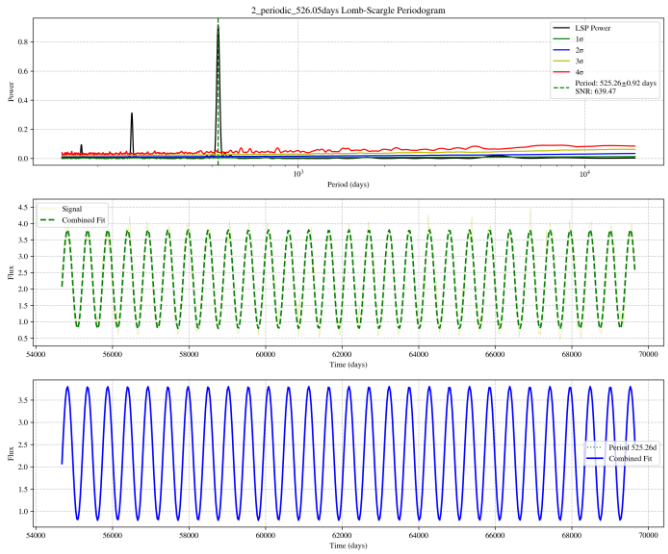


WWZ

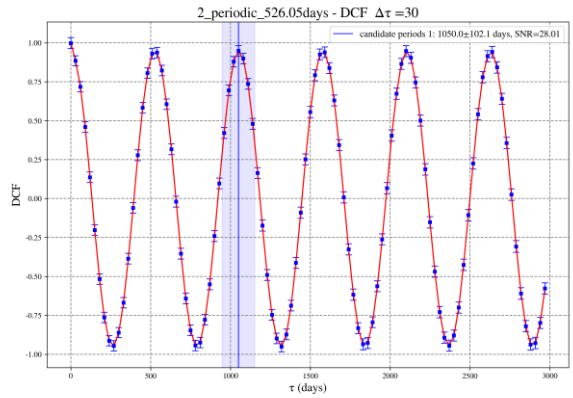
Detailed Information

Period 1: d 1: 245.8128 ± 2.5236 days, WWZ method validation cycle successful：244.6661,LSP Method detected 1 cycles:[245.81280788177347]+-[2.5236340066710756],DCF Method detected 1 cycles:[1230]+-[50.55234392412672],Jurkecivh Method detected 1 cycles:[740]+-[10.0]

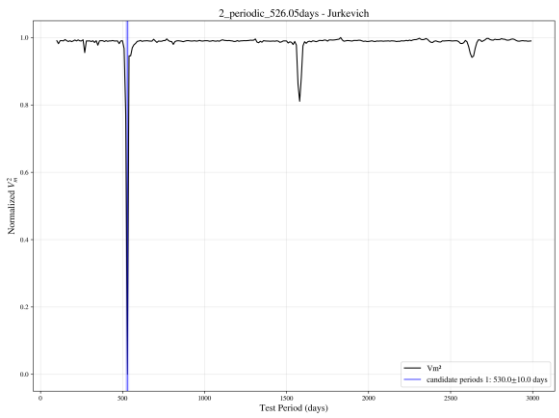
2_periodic526.05days



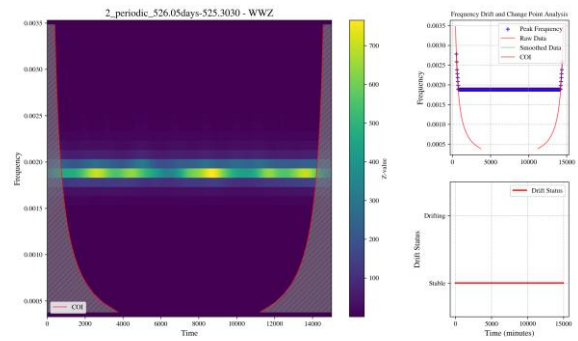
LSP



DCF



JV

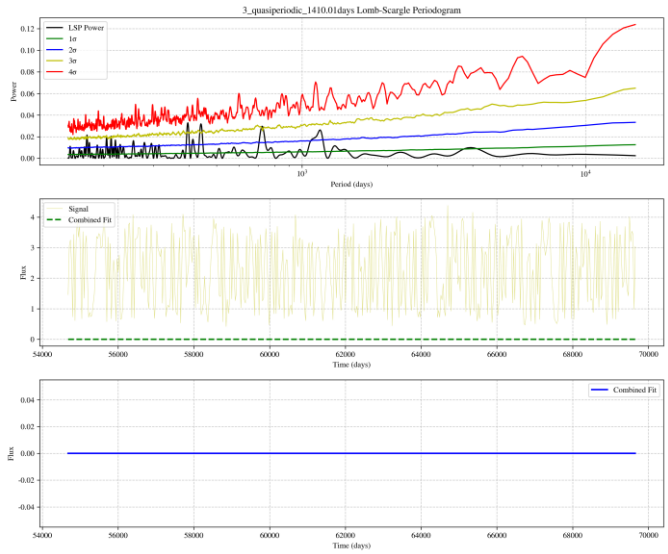


WWZ

Detailed Information

Period 1: d 1: 525.303 ± 0.918 days, WWZ method did not detect this cycle,LSP Method detected 1 cycles:[525.263157894737]+-[0.9215143120959625],DCF Method detected 1 cycles:[1050]+-[102.1000950468279],Jurkecivh Method detected 1 cycles:[530]+-[10.0]

3_quasiperiodic1410.01days



LSP

No DCF Image

DCF

No JV Image

JV

No WWZ Image

WWZ

Detailed Information

Period 1: LSP Method did not detect a cycle, skip this source

4_quasiperiodic1438.46days

LSP

No DCF Image

DCF

No JV Image

JV

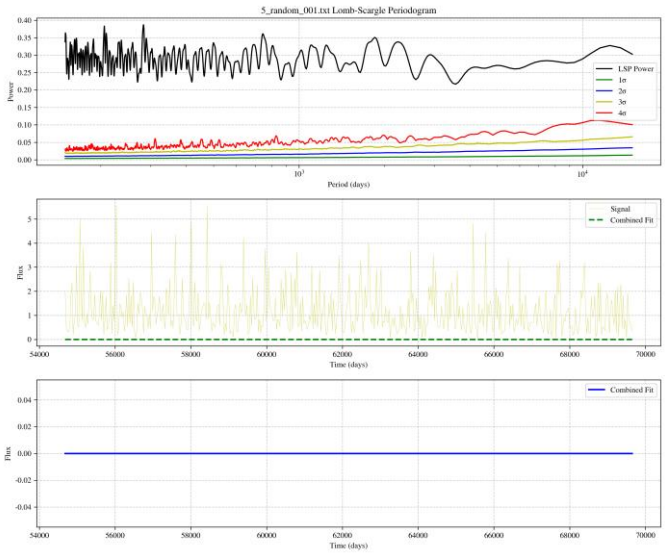
No WWZ Image

WWZ

Detailed Information

Period 1: LSP Method did not detect a cycle, skip this source

5_random001.txt



LSP

No DCF Image

DCF

No JV Image

JV

No WWZ Image

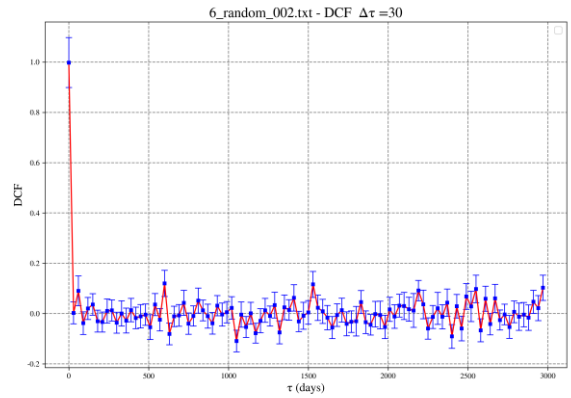
WWZ

Detailed Information

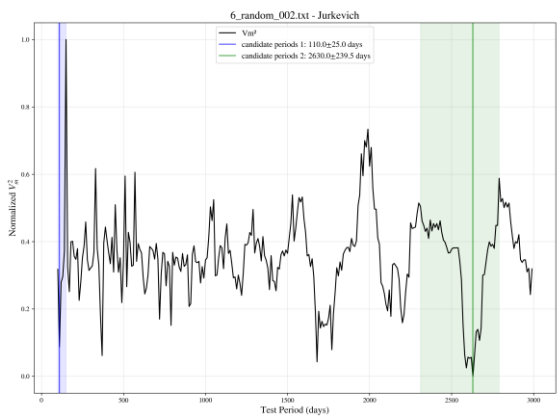
Period 1: LSP Method did not detect a cycle, skip this source

6_random002.txt

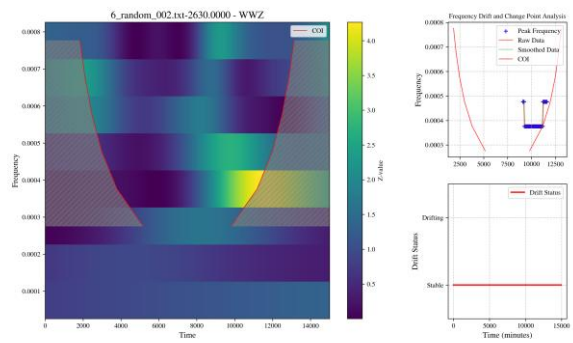
LSP



DCF



JV



WWZ

Detailed Information

Period 1: d 1: 110.0 ± 25.0 days, WWZ method validation cycle successful：120.3150,LSP Method detected 1 cycles:[264.0211640211641]+-[3.3324140251879997],DCF Method did not detect cycle,Jurkecivh Method detected 2 cycles:[110, 2630]+-[25.0, 239.47467705007512]

Period 2: d 2: 2630.0 ± 239.4747 days, WWZ method validation cycle successful：2659.2518,LSP Method detected 1 cycles:[264.0211640211641]+-[3.3324140251879997],DCF Method did not detect cycle,Jurkecivh Method detected 2 cycles:[110, 2630]+-[25.0, 239.47467705007512]